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How Does IT Ambidexterity Impact Organizational Agility?

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Organizational agility is a significant business capability. Though there have been numerous studies about the effects of information technology (IT) capabilities on organizational agility, there has been limited attention on the enabling effects of IT ambidexterity, namely, the dual capacity to explore and exploit IT resources and practices. We propose that IT ambidexterity enhances organizational agility by facilitating operational ambidexterity, and that the magnitude of facilitation depends on the level of environmental dynamism. We test these relationships utilizing data from a large-scale, matched-pair field survey of business and IT executives. The results confirm that a firm's IT ambidexterity does enhance its organizational agility through the mediated effects of operational ambidexterity, and that the dynamism of a firm's environment affects these relationships.

Keywords: agility; IT ambidexterity; operational ambidexterity; environmental dynamism; moderated-mediation analysis

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Introduction

In today's competitive business landscape, *organizational agility* is becoming vital for competitive success because it reflects a firm's ability to continually sense and respond to market changes (D'Aveni et al. 2010, Overby et al. 2006). It has also been recognized that information technologies (ITs) provide a key platform on which contemporary firms compete through utilizing their digitized processes, knowledge, and design capital (Sambamurthy et al. 2003). In line with this, the strategic management of organizational IT resources has attracted attention as a substantial enabler of organizational agility (Bharadwaj 2000). Particularly, with continued advances in IT solutions and services, *IT ambidexterity*—the ability of firms to simultaneously explore new IT resources and practices (IT exploration) as well as exploit their current IT resources and practices (IT exploitation)—has been recognized as a fundamental capability that can enable organizational agility and firm performance (Lee et al. 2007, Subramani 2004). For example, Merrill Lynch's utility model simultaneously leveraged both IT exploration and exploitation for its cost-effective yet flexible IT service provisioning (Levinson 2004).

Whereas Merrill Lynch efficiently utilized its existing IT resources by centralizing them, it implemented an innovative service model based on the actual IT service consumption of other departments. This service model was a disruptive departure from the traditional cost-office IT service model in its industry at that time. Merrill Lynch's ability to simultaneously pursue both IT exploration and exploitation was demonstrated to bring agility into its business processes (Levinson 2004). Such dual capacity in the exploitation and exploration of key organizational resources is known to allow firms' flexible responses to market changes (Gibson and Birkinshaw 2004). However, it remains unclear *how* the ambidextrous use of IT resources leads to greater organizational agility.

Prior studies on the impacts of ITs on organizational agility have examined various IT-based resources and capabilities as enablers of agility. In particular, IT infrastructure has been evaluated as a facilitator of organizational sensing and responsiveness by improving a firm's ability to detect, process, and communicate emerging market information (Chakravarthy et al. 2013, Lu and Ramamurthy 2011). IT management capabilities, such as technical and managerial IT skills

and strategic IT alignment, also contribute to organizational agility by enabling successful implementation or deployment of requisite information systems (Chakravarthy et al. 2013, Lu and Ramamurthy 2011) and by facilitating business-IT collaborations (Tallon and Pinsonneault 2011). Although prior studies in this stream of research do provide some evidence about the positive impacts of various IT resources and capabilities on organizational agility, there are two gaps in the extant literature.

First, prior studies view ITs as valuable and idiosyncratic organizational resources that can lead to higher-level dynamic capabilities, such as agility (Bharadwaj 2000, Sambamurthy et al. 2003). However, the intermediating capability-building processes have seldom been examined with respect to the application of ITs to key business processes, such as operations and marketing, that could enable firms' competitive maneuvers (Grant 1996). Although prior studies have examined certain internal relationships, such as the interactions between IT alignment and IT infrastructure flexibility in volatile environments (Tallon and Pinsonneault 2011) and between IT capabilities and IT spending (Lu and Ramamurthy 2011), they have focused only on IT-related factors; how these interactions translate into agile firms through organizational capability-building processes has not been well understood.

Second, although the ambidextrous management of IT resources and practices could become more vital to the agile maneuvers of contemporary firms, the extant literature has seldom examined IT ambidexterity as a key enabler of organizational agility. Instead, prior studies have either defined IT factors as general latent capabilities or competencies that involve various IT assets, skills, and managerial stances (Chakravarthy et al. 2013, Lu and Ramamurthy 2011), or have focused only on a specific aspect of IT resources, e.g., flexibility of IT infrastructure (Tallon and Pinsonneault 2011). Although some prior studies have articulated the criticality of organizational approaches for both IT exploitation and exploration, they have never explicitly conceptualized or tested the dual capacity of the two distinct IT management approaches (Tallon and Pinsonneault 2011).

The goal of our research is to address these theoretical gaps. First, our research examines a firm's capability-building processes facilitated by IT ambidexterity, which lead to organizational agility. Second, we also examine the specific environmental conditions of the underlying mechanisms for IT-enabled agility creation. We attempt to answer the following key research questions: (1) How does IT ambidexterity enhance organizational agility within a firm? (2) How does the firm's business environment affect

the capability-building processes between IT ambidexterity and organizational agility?

To address the first question, we conceptualize another form of ambidexterity, namely, operational ambidexterity, or the ability to simultaneously pursue operational exploration and exploitation, as an important intermediate process-based capability. To address the second question, our research proposes the conditional role of environmental dynamism (Dess and Beard 1984). With these conceptual developments, we argue that IT ambidexterity enhances organizational agility because it facilitates operational ambidexterity, particularly in highly dynamic environments. We propose two theory-based hypotheses and test them using data gathered through a large-scale, matched-pair field survey of business and IT executives. Since our hypotheses involve both the mediating relationships among organizational capabilities and the conditional effects of environmental dynamism, we utilize an approach that combines mediation and moderation analyses to test our hypotheses (Edwards and Lambert 2007, Muller et al. 2005). Our results indicate that IT ambidexterity enhances organizational agility by improving a firm's operational ambidexterity. Moreover, the mediating effects of operational ambidexterity are more significant in highly dynamic environments. Our research extends the extant literature by revealing the more comprehensive underlying mechanisms between IT ambidexterity and organizational agility.

The rest of this paper is structured as follows. In the next section, we describe the development of our theoretical perspectives and the study's hypotheses. Subsequently, we present the details of our data gathering methodology and analysis. Finally, we discuss the implications of this study for future research and practice.

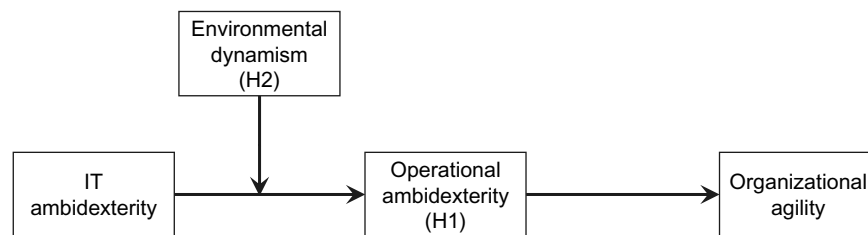
Theory Development

Figure 1 illustrates the conceptual model guiding our research. The logic for this model is influenced by the organizational capability-building and ambidexterity perspectives. We describe the elements of our conceptual development in the following sections.

Capability-Building Processes for Organizational Agility

The strategy literature identifies a firm's capability-building processes as an important mechanism for achieving sustainable competitive advantage (Makadok 2001). Capability-building processes refer to the means by which firms integrate, build, and reconfigure internal and external resources to create organizational dynamic capabilities (Teece et al. 1997). The capability-building processes result in a hierarchy of organizational capabilities in which lower-order

Figure 1 Research Model



functional capabilities are integrated to form higher-order dynamic capabilities (Grant 1991, 1996). These capability-building and integration processes help firms continually develop new competitive actions and gain sustainable competitive advantage over time (D'Aveni et al. 2010).

Consistent with these conceptualizations, information systems (IS) researchers have examined the effects of IT resources and capabilities on organizational agility and firm performance (refer to Appendix A). In this stream of research, organizational agility has been conceptualized as a firm's higher-order dynamic capability to detect opportunities and threats, assemble the needed assets and capabilities to launch an appropriate response, judge the benefits and risks of initiating an action, and execute actions with competitive speed and success (Sambamurthy et al. 2003). ITs are considered an enabler of this higher-order dynamic capability (Chakravarthy et al. 2013, Lu and Ramamurthy 2011). In particular, Sambamurthy et al. (2003) argue that today's digitized platforms of business processes and knowledge allow a firm to rapidly monitor and detect changes, flexibly alter their market strategies, and react more quickly to customers' changing requirements. Thus, a firm's capability to manage and utilize IT resources is vital for organizational agility. According to the capability-building perspective, IT management capabilities are seen as lower-order functional capabilities that are integrated into higher-order capabilities in conjunction with other lower-order functional capabilities embedded in business processes (Grant 1996, Sambamurthy et al. 2003). However, current studies in this stream of research have seldom examined how a firm's IT management capabilities translate into agility through other process-based capabilities.

To address this theoretical gap, we focus first on operational capabilities as key process-based functional capabilities that could be integrated with IT management capabilities to form organizational agility. Operational capabilities represent a firm's ability to perform functional activities through a variety of routines (e.g., selling or manufacturing). These capabilities have been found to enable and facilitate higher-order dynamic capabilities to configure and reconfigure

organizational resources in response to the environment or emerging competitive realities, i.e., the concept of agility in our research (Collis 1994, Pavlou and El Sawy 2010). Second, to provide a comprehensive understanding of how today's diversified IT management activities lead to organizational agility within a firm, we extend the concepts of IT management and operational capabilities from the perspective of ambidexterity—the dual capacity to pursue two disparate things at the same time (Tushman and O'Reilly 1996).

IT and Operational Ambidexterity

Organizational learning theorists identify two distinct means by which firms leverage their resources and capabilities: *exploration* and *exploitation* (e.g., Levinthal and March 1993, March 1991). Exploration refers to the efforts to learn and discover how to combine and recombine potential resources in novel ways to create new capabilities and opportunities, whereas exploitation relates to the efficient leverage and refinement of existing resources through known processes (March 1991). Recent research highlights the synergistic value of the simultaneous pursuit of these two fundamentally different processes, which is known as *ambidexterity* (Gibson and Birkinshaw 2004, Im and Rai 2008). Ambidexterity is argued to be highly desirable for modern firms because they face frequent changes in business environments (Tushman and O'Reilly 1996). To be responsive to environmental changes, firms should be able to conduct both exploration and exploitation in a way that both complement each other and are balanced, rather than increase the tension between the two (Gibson and Birkinshaw 2004, Raisch and Birkinshaw 2008). However, this dual capacity has seldom been considered in the organizational capability-building processes.

By integrating the ambidexterity and capability-building perspectives, we define IT ambidexterity as a firm's ability to simultaneously pursue exploration and exploitation in their management of IT resources and practices, and propose that IT ambidexterity (as a lower-order functional capability) would be an antecedent of organizational agility (a higher-order dynamic capability). IT exploration refers to

the experimentation with new IT resources and practices, whereas IT exploitation refers to the utilization of existing and proven IT resources and practices. IT exploration is associated with the ability of a firm to devote requisite resources and managerial time to gain an understanding of different ITs, experiment with the most promising ones to learn about their functionalities, and select those few technologies that are most likely to have a positive impact on the current and future business operations. In contrast, IT exploitation is associated with the ability of a firm to manage its current portfolio of IT assets, leverage and reuse them in different business activities, invest in complementary technologies to improve their effectiveness, and integrate the technologies with the firm's business processes.

IT ambidexterity is thus a firm's ability to manage both current and emerging technologies, grasp their potential value for business operations, select promising technologies, and implement an efficient means of leveraging those technologies. Seen from the ambidexterity perspective, whereas IT exploration is needed to direct attention toward the emerging technologies, methodologies, and skills, IT exploitation is also needed to ensure that firms are able to fully utilize and leverage their existing IT resources to reap their full benefits. By finding a balance between these goals, firms can better respond to both current and emerging business needs, thus becoming agile.

Likewise, operational ambidexterity refers to a firm's ability to simultaneously pursue operational exploration and exploitation (as another lower-order functional capability). Hammer (2004) proposes two distinctive operational goals and initiatives for competitive success and superior performance, i.e., operational innovation for invention and operational improvement for excellence. Consistent with these conceptualizations, we define operational exploration as inventing new operational techniques or operational processes and operational exploitation as improving current operations by increasing efficiency. Operational exploration is the ability of a firm to fundamentally change or invent new business operations, such as for product/service development and production, supply chain management, or customer delivery or employee management, to create new ways of performing daily tasks (Weerawardena 2003). Operational exploitation is the ability of a firm to enhance operational productivity by improving efficiency, thus reducing the cycle time of current operations and subsequently reducing costs and errors but without fundamentally changing how the operational tasks are accomplished (Chattopadhyay et al. 2001). Specific techniques, such as total quality management and Six Sigma (Powell 1995), are among the methods used for these operational improvements. Therefore, operational ambidexterity relates to a firm's dual

capacity to simultaneously pursue not only the implementation of disruptive ideas for its products and services, production processes, and relevant managerial/administrative activities but also the enhancement of the firm's current operational speed, reliability, and cost.

Hypotheses

IT ambidexterity improves operational ambidexterity by facilitating diverse operational processes within a firm. The use of ITs can enhance the effectiveness and efficiency of business operations, such as better managing customer and supplier relationships (Hammer 2004). Well-applied ITs allow firms to capture information about their customers and marketplaces, share this information among important stakeholders through various communication channels, and facilitate the design, production, and distribution of new products and services (Nambisan 2002). Administrative operations, such as accounting, human resources, and productivity management, are also enhanced by better information processing enabled by the development of appropriate IT resources and practices (Mithas et al. 2011). To initiate and execute such diverse operational activities, firms should provide timely and optimal IT support and infrastructure through modification, development, and/or acquisition of requisite applications, networks, database, and a common IT platform.

Balancing IT exploration and exploitation is particularly essential not only for the effective deployment of potential and existing IT resources and practices but also for the seamless integration and evolutionary adjustments in IT for business processes. For example, Progressive Insurance explored a new information processing environment for its mobile offices that were fully equipped with mobile technologies, and also exploited existing IT infrastructures, including its customer database and corporate network. This, in turn, enabled the firm to achieve superior operational performance. In particular, both IT exploration and exploitation were used for optimizing the firm's service provisioning during the onsite investigation and claim process, which was not only an invention to the insurance industry but also an improvement of the firm's ongoing customer relationship management (Hammer 2004). As a further example, firms' adoption of radio frequency identification technology is known to enable both operational innovation and improvement in areas such as inbound logistics, inventory management, manufacturing, outbound logistics, sales, and customer service. Although a firm's mere adoption of such new technology may not always lead to operational enhancements, the continuous effort to seamlessly integrate existing IT infrastructure and applications

with this new IT initiative for optimizing its information processing will facilitate competitive operational transformations (Piccoli and Ives 2005). Therefore, a firm's dual capacity for IT exploration and exploitation, i.e., IT ambidexterity, is vital to pursue both operational exploration and exploitation for its innovative redesign of operational processes and continuous productivity improvement simultaneously, i.e., operational ambidexterity.

In turn, operational ambidexterity is a critical antecedent of organizational agility. Higher operational capabilities allow firms to effectively and efficiently respond to market competition and changes (Hammer 2004). In particular, exploration for operational innovation enables firms to implement new models of customer service, sales, or manufacturing. Disruptive innovations for products, services, and business processes have been discussed as the driving forces of a firm's agile responses to market changes and opportunities (Charitou and Markides 2003). Likewise, operational exploitation for productivity improvement efforts can support a firm's reconfiguration initiatives to achieve better responsiveness to emerging threats, such as natural disasters or disruptions in the supply chain (Sheffi and Rice Jr 2005). Operational exploitation also increases a firm's ability to take advantage of emerging market opportunities by enabling continuous adaptations (Davenport 1993, Rindova and Kotha 2001). A firm's simultaneous or balanced pursuit of both operational exploration and exploitation, i.e., operational ambidexterity, increases its agility by providing a greater level of flexibility in responding to its market complexity and diversity.

Given as an example of utilizing both operational innovation and improvement, Progressive Insurance could achieve dramatic growth in its highly competitive industry in North America (Hammer 2004). Its new onsite accident investigation policy and practice rapidly induced high customer satisfaction, dramatically reducing claim processing time and costs. At the same time, continuous improvement of the claim process, such as the case-based assignment of an adjuster instead of the immediate assignment of any available agent, also contributed to high customer satisfaction and better service experiences. Progressive Insurance's dual capacity to pursue both operational exploration and exploitation led to better sensing and higher responsiveness to customer needs and preferences, i.e., organizational agility and superior performance. Contrarily, a firm's imbalanced approach can limit its flexible responses to market changes by trapping the firm in specific behavioral patterns (Gibson and Birkinshaw 2004, Levinthal and March 1993). Therefore, a higher level of operational ambidexterity is vital to achieving organizational agility. Based on these arguments, we propose our first hypothesis

about the links between IT ambidexterity and organizational agility:

HYPOTHESIS 1 (H1). *IT ambidexterity enhances organizational agility by facilitating greater operational ambidexterity.*

The hypothesized capability-building processes should be contingent on a firm's environmental context. Environmental context affects the implementation of business strategy; no strategy is universally superior, irrespective of its environmental context (Venkatraman 1989). In particular, environmental dynamism represents the unpredictability and speed of change that a firm must deal with. Lower levels of dynamism are characterized by enduring technologies and stable and predictable consumer preferences. Firms that operate in such environments can produce and sell homogeneous products in large volumes and achieve economies of scale (Johnson et al. 2003). Under these environments, firms can leverage more stabilized and well-practiced routines of action (Ling et al. 2012). Their competitive actions are characterized by fewer and less complex moves.

In contrast, with higher levels of environmental dynamism, firms face significant unpredictability regarding their customers' tastes and preferences, as well as production and service technologies (Bourgeois and Eisenhardt 1988). Firms that operate in such environments need to engage in continual competitive actions, each one of which might confer only short-term or limited competitive advantage (D'Aveni et al. 2010). Under these environments, business and functional competencies might be more necessary so that managers can actively sense opportunities for competitive action and either launch disruptive innovations or protect their firms' interests from rivals' actions (Hammer 2004, Rindova and Kotha 2001).

Consistent with these arguments, in the highly dynamic environments, firms' IT initiatives should respond to emerging and fast-changing business needs by facilitating both their operational innovation and improvement (Chakravarthy et al. 2013, Rindova and Kotha 2001). As the level of environmental dynamism increases, a firm is more likely to need the dual capacity of exploration and exploitation for its functional capabilities to be agile (Jansen et al. 2006, Levinthal and March 1993). IT ambidexterity is more desirable in such environments where firms have higher pressure to build the diversified and flexible operations. Under the increased pressure in such environments, firms that have the dual capacity to deploy requisite IT resources and practices by taking advantage of existing resources and/or experimenting with new technologies will have greater freedom to pursue ambidexterity in planning and managing operational activities.

Conversely, in less dynamic environments, firms' approaches to their markets are simpler, and their competitive actions are less frequent. Thus, the value of the capability-building processes to achieve agility will be reduced. In particular, in less dynamic environments, the relationship between IT ambidexterity and operational ambidexterity will be weaker than that in highly dynamic environments because the need to achieve the dual capacity for their operational ambidexterity is lower. Therefore, the IT-facilitated capability-building processes for organizational agility creation are potentiated under the condition of higher environmental dynamism. Based on these arguments, we develop our second hypothesis about the conditional effect of environmental dynamism on the relationship between IT ambidexterity and organizational agility:

HYPOTHESIS 2 (H2). *An increase in environmental dynamism leads to greater organizational agility because it strengthens the effect of IT ambidexterity on operational ambidexterity.*

Research Method

Data for the study were gathered through a cross-sectional, matched-pair field survey that was complemented by archival data from secondary sources. The population frame for the survey was firms located in China. These firms demonstrated a large variance in terms of their IT and operational maturity. Hence, we believe that they are appropriate for the testing of our hypotheses.

Survey Procedure

We included a broad range of industries and firms to maximize the generalizability of the findings. However, we avoided traditionally labor-intensive industries, such as toy manufacturing and agricultural industries, to focus our attention on industries that require significant IT support for their business operations. Furthermore, our sample frame was limited to firms of medium to large size, as they are more likely to invest in their operations, IT, and strategic activities. Following the National Bureau of Statistics of China, this research defines a medium- to large-sized firm in China as having more than US\$4.5 million in annual sales or more than US\$6 million in total assets.¹

¹ The National Bureau of Statistics of China provides various criteria in defining the standard firm size (i.e., large/medium/small) for different industries. They include annual production capacity, fixed assets, and overall productivity (http://www.stats.gov.cn/english/ClassificationsMethods/Classifications/200210/t20021016_72367.html). We converted the original renminbi to U.S. dollars (USD 1 = CNY 6.67).

The study instruments were pilot tested to ensure their clarity and verify that the questions accurately captured the desired information. Two questionnaires were developed for the study. Whereas the questionnaire used for examining a firm's IT ambidexterity was sent to IT executives (e.g., chief information officers and chief technical officers), the questionnaire designed to study business-related capabilities—organizational agility and operational ambidexterity—and environmental dynamism was sent to business executives (e.g., chief executive officers and chief operating officers). Dual-respondent, matched-pair data were used to reduce the threat of common method bias (Podsakoff and Organ 1986). In addition, secondary data were gathered for some of the control variables, such as total assets and industry characteristics of the sample firms.

To increase the response rate and the respondents' understanding of the survey questions, the survey was conducted using both telephone and face-to-face interviews. Around 1,000 firms located close to three major cities in China—Beijing, Shanghai, and Guangzhou—were randomly selected from the industry directories within each region. After initial contacts were independently made with the business and IT executives of the target firms through mail and telephone, face-to-face interviews were arranged with 290 firms. After removing incomplete data, outliers, and responses from executives outside the target industries, these interviews yielded a sample with 178 matched-pair observations. This represents a response rate of approximately 22%. Although this response rate is deemed reasonably acceptable as a firm-level study, to detect a potential effect of nonresponse bias, patterns of "early" and "late" respondents were compared for the dimensions of firm age, employee number, organizational structure, and industry type (manufacturing versus service). Our independent *t*-tests for these dimensions showed no significant differences between the two groups, suggesting no bias due to nonresponse (Segars 1997).

The final data represents 23 Chinese industry codes at the two-digit level. The respondents' company size varies from 120 to 21,500 employees. The firms sampled include shareholding enterprises (104), foreign-invested enterprises (43), private enterprises (20), and state-owned enterprises (11). Table 1 shows summary information from the firms sampled.

Measurement Development

Our measures were developed in five stages: (1) operationalization of research constructs, (2) item development, (3) validity tests, (4) expert review, and (5) translation from English to Chinese. First, the research constructs were operationalized through definitions of the constructs. Organizational agility was

Table 1 Demographics of Final Samples

Industry name	<i>N</i>	%	Average no. employees	Average total assets (Million US\$)	Average total sales (Million US\$)
Machinery	37	20.8	2,577	234.5	248.4
IT product and service	33	18.5	1,315	237.3	267.3
Electronics	25	14.1	1,652	172.5	132.9
Chemical products	14	7.9	2,260	312.8	243.6
Medicine and biological products	13	7.3	1,843	333.3	309.3
Metal and nonmetal	10	5.6	3,001	1,456.7	1,259.3
Textile and apparel	10	5.6	1,951	177.8	175.4
Wholesale	5	2.8	755	35	166.1
Conglomerate	5	2.8	1,615	426.6	1,022
Others	26	14.6	1,767	354.9	283.2
Total or average (sample level)	178	100	1,927	322.5	317
SD (sample level)			2,869	1,059	969.2
Average (industry level)			1,874	374.1	410.7
SD (industry level)			633	396.2	392.7

Notes. These data were compiled over the years 2003–2005. SD, Standard deviation.

operationalized as a second-order latent construct having four reflective first-order constructs. Consistent with the work of other researchers (Gibson and Birkinshaw 2004, Im and Rai 2008), the two ambidexterity constructs, i.e., IT ambidexterity and operational ambidexterity, were operationalized using item-level interaction terms of their subdimensions. For environmental dynamism, we utilized perceived environment turbulence as a firm-level continuous measurement consisting of multiple items. Table 2 summarizes the operational definitions of the principal research constructs according to their first-order constructs or subdimensions.

We also included control variables in the form of organizational and industrial characteristics to control for their potential impact on both the mediating and dependent variables as recommended by Preacher et al. (2007). These include firm size, firm age, organizational types, structures, and industry growth.

Every attempt was made to use existing validated measures that have good psychometric properties, although we made some modifications to suit the context of our research. However, in cases where there were no appropriate preexisting measures, we developed new measures (refer to Appendix B for the final items). All of the measurement items were based on a seven-point Likert scale with 1 equated to very weak or strongly disagree and 7 equated to very strong or strongly agree.

To establish construct validity of the adapted and self-developed items, the procedure described by Moore and Benbasat (1991) was carried out. This included three rounds of a structured sorting involving academic faculty (five), doctoral students (four), master's students (three), and industry managers (three) from the IS and management areas. In each round, five judges were employed to sort the items based on the given definitions of research constructs.

Necessary changes were then made to the survey instrument. The content validity of the items was assessed by three senior IS researchers, who have expertise in the specific research area of IT strategy and management. Their comments and suggestions were used to further improve the two surveys.

Finally, using a translation committee approach, i.e., a committee of bilingual participants, the original English instrument was translated to Chinese specifically for data collection in China. This committee approach is considered useful for establishing psychological as well as linguistic equivalence via the sense-making process taking place among committee members (van de Vijver and Leung 1997). Four native Chinese speakers fluent in English and knowledgeable about the subject matter formed the translation committee.

Results

Conceptually, our hypotheses combined a mediation effect that is the impact of IT ambidexterity on organizational agility through operational ambidexterity (H1) and a moderation effect from the contingency effect of environmental dynamism on the mediating process (H2). This is consistent with the moderated-mediation perspective (Edwards and Lambert 2007, Muller et al. 2005).² The salient question

² There are two perspectives on combining mediation and moderation effects, i.e., mediated moderation and moderated mediation (Edwards and Lambert 2007, Muller et al. 2005). The mediated-moderation perspective builds on a conceptual argument about a moderation effect on an overall treatment effect and asks if a mediating process is responsible for that moderation (Muller et al. 2005, p. 853). Based on this perspective, Tiwana and Konsynski (2010) theorize and validate that the interaction between IT architecture modularity (treatment) and IT governance decentralization (moderator) influences on IT alignment (outcome) through IT agility

Table 2 Operational Definitions of the Principal Research Constructs

Constructs	Operational definitions
Organizational agility	Consistent with Bharadwaj and Sambamurthy (2005) and Lu and Ramamurthy (2011), agility was operationalized as a higher-order dynamic capability having multiple facets to effectively and efficiently sense and respond to various market conditions. The following four reflective first-order constructs were measured for organizational agility: • Proactiveness: The ability to capture new opportunities ahead of competitors by anticipating and responding proactively (Lumpkin and Dess 1996, Miller and Friesen 1983) • Radicalness: The ability to initiate radical strategic movements by implementing new business models to penetrate new markets (Miller and Friesen 1983, Zahra and Covin 1995) • Responsiveness: The ability to rapidly respond to emerging opportunities in the marketplace due to changes in customer preferences or the environment (Hult et al. 2005, Tracey et al. 1999) • Adaptiveness: The ability to adapt business models and keep up with emerging industry best practices (Jarrar and Zairi 2000, Rindova and Kotha 2001, Subramani and Youndt 2005)
IT ambidexterity	IT ambidexterity is the ability of a firm to manage its attention to both the current and emerging technologies, understand their potential value for business operations, select promising technologies, and implement an efficient means of leveraging those technologies. Following Gibson and Birkinshaw (2004), IT ambidexterity was operationalized using the item-level interaction terms of IT exploration and IT exploitation capabilities. • IT exploration capability: An ability to acquire and experiment with new IT resources and practices (e.g., IT architecture, IT applications, IT skills, and IT development methodologies) (Nambisan et al. 1999) • IT exploitation capability: An ability to utilize existing IT resources and practices (Atuahene-Gima 2005, Ravichandran and Lertwongsatien 2005)
Operational ambidexterity	Operational ambidexterity is the ability of a firm to continually innovate and improve its operational processes. Following Gibson and Birkinshaw (2004), operational ambidexterity was operationalized using the item-level interaction terms of operational exploration and operational exploitation capabilities. • Operational exploration capability: An ability to fundamentally change or invent new business operations (e.g., product/service development and production, supply chain management, customer delivery, and employee management) to create new ways of performing daily tasks (Hammer 2004, Weerawardena 2003) • Operational exploitation capability: The ability to enhance operational productivity by improving the efficiency and cycle time of current operations and reducing their cost (Chattopadhyay et al. 2001, Hammer 2004)
Environmental dynamism	Environmental dynamism is the unpredictability of contextual changes that a firm has to deal with (Eisenhardt and Martin 2000). This construct was operationalized as a firm's perceived environmental turbulence due to rapid changes in its business environment (Atuahene-Gima 2005, Wade and Hulland 2004).

of the moderated-mediation perspective is whether the mediated effect varies as a function of a moderator ([Muller et al. 2005](#)). To test these moderated-mediation hypotheses, the ordinary least squares (OLS) analysis was conducted ([Edwards and Lambert 2007](#), [Muller et al. 2005](#)) using SPSS version 20. A covariance-based structural equation modeling technique, EQS version 6.1, was used to further check the robustness of our measurement and path models ([Bentler 2004](#)).

For the OLS analysis, we calculated a single indicator for each of our principal constructs using the latent factor score approach. This latent factor approach is recommended for calculating single measures for research variables having multiple indicators; in this approach, the multiple indicators are weighted to resolve the potential biasing effects of measurement error ([Muller et al. 2005](#)). Consistent with [Yuan et al. \(1997\)](#), three stages were utilized to develop and evaluate the measurement model. First, we conducted validity tests for the original measures of the directly observed research constructs that are

the first-order constructs of organizational agility and the subdimensions of IT and operational ambidexterity. The purpose of this step was to assess whether or not they had valid factor structures for the proposed conceptual model. Next, we developed new indicators for our principal research constructs. To develop new indicators of agility, the latent factor scores of its first-order constructs were calculated. For IT and operational ambidexterity constructs, their item-level interaction terms were calculated. Last, we repeated the latent factor score approach and calculated the single indicator of each of the principal constructs for our hypothesis tests that used OLS regression analysis.

Measurement Model Evaluation

For this evaluation, the psychometric properties of our original instrument were first assessed. Our approach included testing to assess (1) item reliability, (2) construct reliability, and (3) measurement validity ([Hair et al. 1998](#)). First, all of the factor loadings were above the acceptable threshold (0.70). Second, the Cronbach's alpha scores for all of the constructs consisting of multiple indicators were higher than the recommended threshold (0.70). These results confirmed that all of the reflective multi-indicator constructs had satisfactory reliability (refer to Appendix C for details

(mediator). On the other hand, the moderated mediation takes place when a mediating process is responsible for producing a treatment effect and the magnitude of the mediating process varies as a function of a moderator variable ([Muller et al. 2005](#), p. 854), as in our case.

Table 3 Correlation Matrix Among Directly Observed Constructs

Constructs	Means (SD)	<i>PRO</i>	<i>RAD</i>	<i>RES</i>	<i>ADP</i>	<i>OERC</i>	<i>OEIC</i>	<i>IERC</i>	<i>IEIC</i>	<i>EVD</i>
<i>Proactiveness (PRO)</i>	5.24 (0.72)	0.90								
<i>Radicalness (RAD)</i>	3.91 (1.08)	0.57**	0.90							
<i>Responsiveness (RES)</i>	5.12 (0.74)	0.71**	0.56**	0.91						
<i>Adaptiveness (ADP)</i>	4.37 (0.90)	0.50**	0.59**	0.48**	0.90					
<i>Operational exploration capability (OERC)</i>	4.52 (0.90)	0.47**	0.56**	0.57**	0.48**	0.90				
<i>Operational exploitation capability (OEIC)</i>	4.74 (0.78)	0.47**	0.46**	0.48**	0.60**	0.49**	0.89			
<i>IT exploration capability (IERC)</i>	4.50 (0.83)	0.24**	0.32**	0.33**	0.42**	0.32**	0.40**	0.83		
<i>IT exploitation capability (IEIC)</i>	5.11 (0.82)	0.13	0.20**	0.26**	0.20**	0.21**	0.33**	0.50**	0.84	
<i>Environmental dynamism (EVD)</i>	4.40 (1.19)	0.33**	0.59**	0.45**	0.44**	0.51**	0.41**	0.38**	0.29**	0.84

Notes. The diagonal elements in this correlation matrix are the square root of the average variance extracted. SD, Standard deviation.

** $p < 0.01$.

of the reliability analysis results). Third, the validity of all of the measurements was tested using an explorative factor analysis with principal component analysis (refer to Appendix C for the cross-loadings). All items loaded more highly on the intended construct than on any other constructs, providing evidence of discriminant and convergent validity among the multi-indicator constructs (Hair et al. 1998). Table 3 shows the correlations among the constructs. Table 3 shows moderate correlations among some of the first-order constructs of organizational agility and the subdimensions of operational ambidexterity. In particular, the reflective first-order constructs of organizational agility, specifically proactiveness, radicalness, responsiveness, and adaptiveness, showed relatively high correlations (0.48–0.71); this is because they share the same underlying latent construct of agility (Petter et al. 2007). In addition, the subdimensions of operational ambidexterity also showed moderate correlations with the first-order constructs of organizational agility (0.46–0.60). This is expected because they represent different levels of business capabilities within a firm (Collis 1994). We additionally tested their discriminant validity using the average variance extracted (AVE). Discriminant validity is inferred when the square root of each construct's AVE is higher than the correlation of the construct to other latent variables (Barclay et al. 1995). All of the diagonal values in Table 3, which are the square root of each construct's AVE, were much higher than the construct's correlations with other constructs. The results provide additional evidence of discriminant validity among the multi-indicator constructs.

A confirmatory factor analysis (CFA) was conducted to further assess the overall goodness of fit of

our measurement model using maximum likelihood estimates of EQS version 6.1 (Bentler 2004). The CFA model fit indices of the measurement model were within guidelines, confirming the goodness of fit of our measurement model (Purvis et al. 2001): $\chi^2 = 366$; $df = 286$; incremental fit index, 0.97; nonnormed fit index, 0.97; comparative fit index, 0.97. In addition, root mean-square error of approximation (RMSEA) equaled 0.04; a RMSEA of less than 0.08 is considered to have practical evidence of good model fit (Browne and Cudeck 1993).

Based on these validity results for the directly observed research constructs, organizational agility was formed using the latent factor scores of its four first-order constructs as its new reflective indicators (Chin 1998). For the IT ambidexterity and operational ambidexterity constructs, the item-level interaction terms were calculated by multiplying the three items of IT exploration capability with the three items of IT exploitation capability (for a total of nine indicators) and by multiplying the three items of operational exploration capability with the three items of operational exploitation capability (for a total of nine indicators) (see Gibson and Birkinshaw 2004, He and Wong 2004). Last, using these new indicators, the factor scores of organizational agility and the two ambidexterity constructs were calculated for further hypothesis tests.

Hypothesis Testing

To test the mediated relationships (H1), we conducted Baron and Kenny's (1986) mediation test and Sobel's (1982) standard errors test. To test the moderating effects of environmental dynamism on the mediated relationships (H2), we conducted both the Muller

Table 4 Results of Testing for Mediating Effects

	Baron and Kenny (1986) test				Sobel (1982) test	VAF test
	Step 1 (DV: Agility)	Step 2 (DV: Operational ambidexterity)	Step 3 (DV: Agility)	Step 4 (DV: Agility)	Mediation effects (<i>t</i>)	VAF (<i>z</i>)
IVs						
IT ambidexterity	0.40** (<i>t</i> = 5.44)	0.43** (<i>t</i> = 5.90)		0.13* (<i>t</i> = 2.01)	0.30** (<i>t</i> = 5.35)	0.70** (<i>z</i> = 5.34)
Operational ambidexterity			0.69** (<i>t</i> = 12.38)	0.64** (<i>t</i> = 10.55)		
Controls						
Firm size	−0.12 (<i>t</i> = 1.46)	−0.08 (<i>t</i> = 0.93)	−0.09 (<i>t</i> = 1.34)	−0.07 (<i>t</i> = 1.12)		
Firm age	−0.03 (<i>t</i> = 0.40)	−0.04 (<i>t</i> = 0.64)	0.00 (<i>t</i> = 0.04)	0.00 (<i>t</i> = 0.00)		
Private organization	0.01 (<i>t</i> = 0.09)	−0.05 (<i>t</i> = 0.71)	0.04 (<i>t</i> = 0.64)	0.04 (<i>t</i> = 0.69)		
Listed organization	−0.10 (<i>t</i> = 1.32)	−0.23** (<i>t</i> = 2.99)	0.04 (<i>t</i> = 0.71)	0.04 (<i>t</i> = 0.71)		
Organizational structure	−0.17* (<i>t</i> = 2.55)	−0.14* (<i>t</i> = 2.15)	−0.08 (<i>t</i> = 1.43)	−0.08 (<i>t</i> = 1.52)		
Industry growth	−0.20* (<i>t</i> = 2.69)	−0.13 (<i>t</i> = 1.80)	−0.07 (<i>t</i> = 1.34)	−0.12* (<i>t</i> = 1.98)		
<i>R</i> ²	0.24	0.27	0.53	0.54		
Adjusted <i>R</i> ²	0.20	0.23	0.51	0.52		

p* < 0.05; *p* < 0.01.

et al. (2005) moderated causal steps analysis and Edwards and Lambert's (2007) constrained nonlinear regression analysis.

Mediating Effects (H1) Test. We first tested the hypothesized relationship about mediated effects (H1) by using the four-step procedure suggested by Baron and Kenny (1986). The results are reported in Table 4.

First, as shown in Table 4, there was a significant effect of IT ambidexterity (the independent variable (IV)) on organizational agility (the dependent variable (DV)) without involving operational ambidexterity (the mediating variable (MV)). Second, there was a significant effect of IT ambidexterity (IV) on operational ambidexterity (MV). Third, there was a significant effect of operational ambidexterity (MV) on organizational agility (DV). Finally, in the presence of operational ambidexterity (MV), the effect of IT ambidexterity (IV) on organizational agility (DV) was reduced, though it was still significant. Taken together, the results suggest that operational ambidexterity (MV) acts as a partial mediator.

Next, we conducted the Sobel's (1982) standard errors test to further ascertain the mediating relationship. As shown in Table 4, there was a significant mediation effect of operational ambidexterity (*p* < 0.01). This significant mediation effect was also confirmed (again with *p* < 0.01) by the variance accounted for (VAF) test (Eberl 2010). Our results show the proposed mediation relationships to be significant.³ Therefore, the mediation hypothesis (H1) is supported.⁴

³ To check the robustness of the OLS analysis results, we repeated the same tests using EQS, i.e., a covariance-based structural equation modeling technique, and achieved consistent results.

⁴ In the mediation model tests, most of the control variables did not have significant effects on agility and operational ambidexterity,

Moderating Effects (H2) Test. We tested our moderating effects hypothesis (H2) using the moderated-mediation analysis approaches (Edwards and Lambert 2007, Muller et al. 2005). Following Muller et al. (2005), we tested three distinct models. The first model tested the moderating effects of environmental dynamism on the direct link between IT ambidexterity and agility, which examines the overall treatment effect without involving operational ambidexterity. The second model tested the moderating effects of environmental dynamism on the first-stage mediation link, i.e., the link between IT ambidexterity and operational ambidexterity. The final model tested the moderating effects of environmental dynamism on both the second-stage mediation link, i.e., the link between operational ambidexterity and agility, and the residual direct link. To test the moderation effects of environmental dynamism on the proposed mediation stages, we created the single interaction terms between IT ambidexterity and environmental dynamism and between operational ambidexterity and environmental dynamism using the standardized product-indicator approach (Muller et al. 2005, Tiwana and Konsynski 2010).⁵ Table 5 shows the

although a few firm-level and industry-level characteristics showed significant effects: listed organization type had a negative relationship with operational ambidexterity (Model 2, *p* < 0.01) and organizational structure (product centered versus geographic centered) had negative relationships on agility (Model 1, *p* < 0.05) and operational ambidexterity (Model 2, *p* < 0.05). Industry growth had a negative relationship with agility (Models 1 and 4, *p* < 0.05).

⁵ Using the variance inflation factor (VIF), we tested potential multicollinearity when adding the product terms to the models. The VIF scores of all of the principal constructs including the product terms are between 1.23 and 1.62. The VIFs are below a conservative threshold of 5. These results indicate that our model has low multicollinearity concern (Thatcher and Perrewé 2002).

Table 5 Results of Moderated-Mediation Analysis

Predictors	Model 1. Overall treatment effect (DV: Agility, without mediator)	Model 2. First-stage mediation link (DV: Operational ambidexterity)	Model 3. Second-stage mediation link (DV: Agility, with mediator)
<i>IT ambidexterity</i>	0.18* ($t = 2.48$)	0.23** ($t = 3.15$)	0.06 ($t = 0.90$)
<i>Environmental dynamism</i>	0.46** ($t = 6.83$)	0.41** ($t = 6.10$)	0.24** ($t = 3.77$)
<i>IT ambidexterity</i> × <i>Environmental dynamism</i>	0.16* ($t = 2.41$)	0.13* ($t = 2.05$)	0.10 ($t = 1.66$)
<i>Operational ambidexterity</i>			0.54** ($t = 8.05$)
<i>Operational ambidexterity</i> × <i>Environmental dynamism</i>			−0.04 ($t = 0.62$)

* $p < 0.05$; ** $p < 0.01$.

results of the moderated causal steps analysis proposed by Muller et al. (2005).

As reported in Table 5, the moderating effect of environmental dynamism on the first-stage mediation link between IT ambidexterity and operational ambidexterity (Model 2) was significant ($\beta = 0.13$, $p < 0.05$); correspondingly, the partial treatment effect (i.e., the main effect) of operational ambidexterity on agility in the second-stage mediation link (Model 3) was also significant ($\beta = 0.54$, $p < 0.01$). Together, they indicate that the moderated-mediation effect is not zero (Muller et al. 2005). The results lend support to Hypothesis 2 arguing that under higher environmental dynamisms, due to the greater effects of IT ambidexterity on operational ambidexterity (Model 2), the mediating relationships among IT ambidexterity, operational ambidexterity, and agility will be stronger. However, our moderated-mediation relationships were not found to be a “prototypic” case because the overall treatment (direct) effect of IT ambidexterity on organizational agility (Model 1) was also moderated by the environmental dynamism (Muller et al. 2005).⁶ On the other hand, the moderating effects on the residual direct effect of IT ambidexterity on organizational agility (Model 3) was insignificant, indicating no significant difference of the direct effects under different levels of environmental dynamism in our moderated-mediation model.

⁶ A prototypic case of moderated-mediation means that “what varies as a function of the moderator is not the magnitude of the overall treatment effect on the outcome but the mediating process that produces it” (Muller et al. 2005, p. 854). According to Muller et al. (2005), to set up such a prototypic moderated-mediation case, the following four conditions need to be confirmed: (1) a significant overall treatment effect; (2) no influence of the moderator on this overall treatment effect; (3) significant moderation effects on either the first-stage or second-stage mediation link, or both; and (4) parallel to the third condition, either a significant partial treatment effect (i.e., main effect) in the second-stage mediation link (in the case of significant first-stage moderation) or a significant partial treatment effect in the first-stage mediation link (in the case of significant second-stage moderation), or both.

To investigate the conditional values of the entire mediating path, which is recommended by Edwards and Lambert (2007), we tested the product term of the first-stage and second-stage mediation links, i.e., the mediated effect under different levels of moderator.⁷ For this test, a higher environmental dynamism group (greater than one standard deviation above the mean, $n = 33$) and a lower environmental dynamism group (less than one standard deviation below the mean, $n = 32$) were used. The test results reveal that the mediated effect under high environmental dynamism (0.18, $t = 3.09$) was significant ($p < 0.01$), whereas the mediated effect under low environmental dynamism (0.06, $t = 0.76$) was not. These indicate that the mediated effects in our model depend on the degree of environmental dynamism. Thus, IT ambidexterity affects organizational agility by enhancing operational ambidexterity, more strongly under conditions of higher environmental dynamism, which lends further support for the moderated-mediation effects.

Before we discuss the implications of our findings, it is important to consider the limitations of the research. First, we utilized a cross-sectional research design that naturally limits the study of the causal relationships among the research constructs. Second, the data used in this research were collected within Chinese firms and thus the findings of the study could be subject to the influence of certain characteristics of this specific economy. In a globalized competitive context, most studies are rooted in the North American context. As China emerges as an important economy and a context for intensified global competition, we believe that our unique data set provides an important extension to the repository of research on IT and competitive success. Third, although our research model depicts that a firm’s IT investments

⁷ Following Edwards and Lambert (2007), the product term was divided by the square root of its estimated variance, and the resulting ratio was interpreted as a t -statistic. Since this approach requires the normality assumption, bootstrapping was used to relax the foregoing assumption.

may lead to organizational agility and better performance, we recognize that the enhanced organizational performance and agility may exert reverse feedback on organizational IT resources and capabilities. Organizations can have bidirectional relationships among organizational competitive activities, high-level dynamic capabilities, and IT capabilities and activities, known as the coevolutionary adaptation processes (Sambamurthy et al. 2003). The feedback effects can be explained by the learning cycle of organizational dynamic capabilities (Teece et al. 1997). These potential feedback effects for the proposed relationships could be vital when considering the assumption of a unidirectional relationship taken by the regression and path analyses adopted for our hypothesis tests (Edwards and Lambert 2007).

Discussion

With intensifying levels of rivalry and advancements in the functionalities of ITs, there is a greater interest in examining how the management and use of ITs affects firms' competitiveness. In particular, the impacts of ITs on organizational agility have begun to attract significant theoretical and empirical attentions (e.g., Chakravarthy et al. 2013, Lu and Ramamurthy 2011, Sambamurthy et al. 2003). However, in the extant literature, the intermediating mechanisms and conditions that enable IT-facilitated capability-building processes within a firm have seldom been examined; thus, *how* the management of IT resources and practices affects organizational agility is still under-researched. To address these theoretical gaps, this research draws on the capability-building perspective to theorize that operational capabilities would mediate the links between the management of IT resources and organizational agility. Integrating this view with the ambidexterity perspective, we propose that IT ambidexterity, the ability to simultaneously explore and exploit IT resources and practices, would enable increased organizational agility by facilitating the development of operational ambidexterity, the process-based dual capacity of operational exploration and exploitation. Finally, we propose that the levels of environmental dynamism would moderate this mediated relationship.

The results from our empirical analyses confirm our hypotheses. Specifically, the results confirm that operational ambidexterity significantly mediates the relationship between IT ambidexterity and agility. The results also confirm that environmental dynamism moderates the mediated relationships, because it has a positive moderation effect on the relationship between IT ambidexterity and operational ambidexterity. Consistently, operational ambidexterity fully mediates the relationship between IT ambidexterity and agility at higher levels of environmental

dynamism. In contrast, at lower levels of environmental dynamism, there is no significant mediation effect.

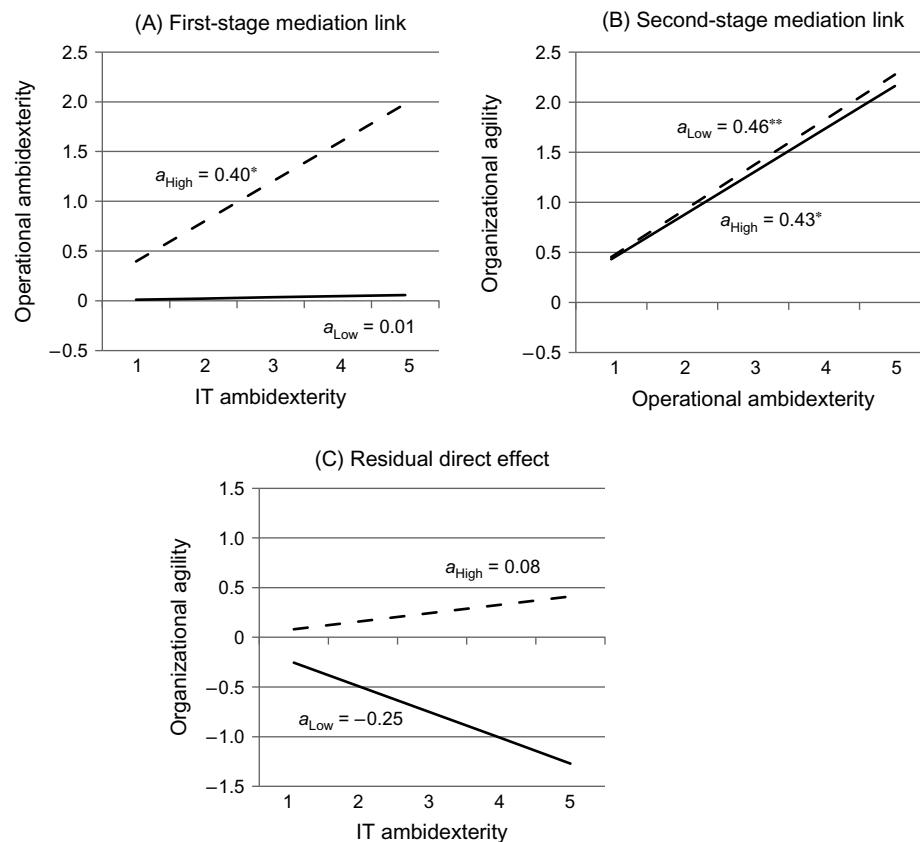
Implications

Our findings have several important implications for research and practice. First, our findings on the significant mediating effect of operational ambidexterity between IT ambidexterity and organizational agility reveal how a firm's lower-order functional capabilities are integrated to form higher-order dynamic capabilities (Grant 1996). In this light, IT management capabilities can be considered to provide a digitized platform that facilitates the building of business process capabilities, such as operational capabilities, which, in turn, enable higher-order dynamic capabilities, such as organizational agility. Moreover, our findings confirm the importance of the ambidextrous management of IT resources and practices to achieve organizational agility. Recent studies highlight the synergistic value of ambidextrous activities pursued by the same business unit, department, or employee, particularly to achieve business flexibility (Gibson and Birkinshaw 2004, Im and Rai 2008). Our findings confirm that IT ambidexterity plays the role of an important enabler of operational ambidexterity, whereas operational ambidexterity is more likely to be a proximal antecedent to agility.

Second, the moderated-mediation findings suggest that the role of operational ambidexterity as a mediator between IT ambidexterity and organizational agility is much more significant in firms that operate under conditions of high environmental dynamism. The greater mediating effects of operational ambidexterity in a highly dynamic environment point to the critical role of process-based functional capabilities in managing the dynamics of continuous and rapid competitive actions in such environments. Moreover, our findings reveal that this increase in mediated effect in those environments is primarily due to the greater facilitating role of IT ambidexterity in improving operational ambidexterity. To further confirm our findings, following Preacher et al. (2007), we also analyzed the plots of simple paths for the first and second-stage mediation links. Figure 2 shows the results.

As shown in these plots, both the first-stage mediation link from IT ambidexterity to operational ambidexterity (plot (A)) and the second-stage mediation link from operational ambidexterity to organizational agility (plot (B)) had steeper slopes when the environmental dynamism was higher (more than one standard deviation above the mean, $n = 33$) than when it was lower (less than one standard deviation below the mean, $n = 32$). However, the difference in the slope for the first-stage mediation link ($\Delta a_{\text{High-Low}} = 0.39$) was much larger than that of the

Figure 2 Plots of Simple Paths and Effects



Notes. Slopes a_{High} and dashed lines indicate high environmental dynamism ($\geq +1$ SD, $n = 33$), and slopes a_{Low} and solid lines indicate low environmental dynamism (≤ -1 SD, $n = 32$).

* $p < 0.05$; ** $p < 0.01$.

second-stage mediation link ($\Delta a_{High-Low} = 0.03$). These findings confirm that the greater mediated effect of the highly dynamic group is mainly derived from the greater effects of IT ambidexterity on operational ambidexterity. These imply that IT ambidexterity is particularly important under a high level of environmental dynamism because it confers a dual capacity to pursue operational innovation and improvement simultaneously.

To further understand IT ambidexterity, it is also useful to investigate the direct effect of IT ambidexterity on agility, even though it is not a main focus of the research. IT resources and capabilities have been discussed to have a direct effect on agility because emerging ITs may directly enhance a firm's capability of sensing and responding to new market opportunities (Overby et al. 2006). Similarly, predefined configurable systems are argued to be useful, especially for foreseen opportunities or threats (Sheffi and Rice Jr 2005), which means that the effects of predefined IT-based solutions may be more useful in less dynamic environments than in highly dynamic environments. These potential conditional effects could be part of

a direct path between IT ambidexterity and organizational agility in our research model. However, our moderated-mediation test results indicate no significant difference of the residual direct effects (i.e., the direct effect with the addition of the mediating variable to the model) under different levels of environmental dynamism. We further examined the plots of simple paths for the residual direct effects under different levels of environmental dynamism. Plot (C) in Figure 2 shows that under the higher levels of dynamism, the residual direct effects showed a positive slope ($a_{High} = 0.08$), whereas under low levels of dynamism, the residual direct effects showed a negative slope ($a_{Low} = -0.25$).

These findings imply that in a highly dynamic environment, a firm's IT ambidexterity has greater impacts on its organizational agility through both the indirect and direct links than it does in a lower dynamism environment. These lend support to the arguments that highlight the importance of both the IT-facilitated indirect and direct effects in developing organizational dynamic capabilities (Chakravarthy et al. 2013).

Finally, our findings also have implications for practitioners to better understand the enhancement of organizational agility. In particular, our findings suggest that IT managers, or top executives, should be aware of the importance of simultaneous IT exploration and exploitation in their organizational contexts to actively promote both IT exploration and exploitation pursuits within their firms to create organizational agility. Organizational exploration and exploitation have been believed to be difficult to attain simultaneously because firms are easily biased in pursuing only one of the two distinct behavioral approaches due to their customs and inertia (March 1991). Therefore, achieving the balanced capabilities of managing IT resources and practices should be a significant business agenda for contemporary firms. Our findings also suggest that contemporary firms should invest in IT along with their operational initiatives. In particular, senior executives should be cognized that appropriate balance between IT exploration and exploitation will enhance the effectiveness of operational initiatives, which, in turn, makes organizations more agile, especially when facing a highly dynamic business environment.

Contributions

The key contribution of our research lies in its theoretical extensions of the extant IT-enabled agility creation literature by providing an advanced nomological model of the relationships among IT management capabilities, operational capabilities, organizational agility, and environmental conditions. First, adopting the organizational capability-building perspective, our model proposes that a firm's IT management and operational capabilities, as the lower-level functional capabilities, are integrated into its organizational agility, a form of higher-level dynamic capabilities (Grant 1996). In particular, our research reveals how IT management activities translate into agile organizations through operational capability-building processes. In the extant literature, these underlying mechanisms of IT-facilitated capability-building through key operational capabilities have seldom been discussed because how ITs can be applied to operational processes has not been examined (e.g., Chakravarthy et al. 2013, Lu and Ramamurthy 2011, Tallon and Pinsonneault 2011). Our theoretical development of this advanced nomological model also involves an extended set of conceptualizations of the key factors by adopting the ambidexterity perspective. As the ambidexterity of a firm has become more significant for flexible responses to market changes in today's competitive business environment, our theoretical development with this emerging perspective can provide a more comprehensive understanding of the role of IT management in contemporary firms.

This theoretical development and its empirical validation address our first research question: *How does IT ambidexterity enhance organizational agility within a firm?*

Second, we further incorporated the conditional effects of environmental dynamism into our nomological model. These conditional effects have seldom been examined in prior IT-enabled agility studies. For example, although Lu and Ramamurthy (2011) found a significant effect of IT capabilities on agility, they did not account for the conditional effects of environmental conditions. Although a few prior studies have examined the conditional values of organizational IT resources and capabilities in leading to organizational agility, these studies did not examine the intermediating mechanisms of agility creation through business processes. Thus, they failed to uncover under what business conditions ITs could be applied to key business processes to enhance organizational agility (e.g., Chakravarthy et al. 2013, Tallon and Pinsonneault 2011). Through our research, these conditional effects were tested using an analysis approach to combine the mediation and moderation effects, i.e., the moderated-mediation analysis. This approach arguably provides a better understanding of how the complex mediating relationships within a nomological model are influenced by the moderator as a contingency factor (Edwards and Lambert 2007, Muller et al. 2005). This theoretical development and its empirical validation address our second research question: *How does the firm's business environment affect the capability-building processes between IT ambidexterity and organizational agility?* Therefore, by clearly identifying the underlying mechanism as well as the boundary condition of IT-enabled capability building, our research significantly advances the current understanding and literature on IT-enabled agility creation.

Future Research

Our conceptualizations and findings can serve as a base for future research. For example, first, our findings show that the mediated effect (or indirect effect) through operational ambidexterity becomes greater in a highly dynamic environment, whereas the residual direct effect between IT ambidexterity and agility has a negative value in a low dynamic environment. This evokes further questions about the alternative roles (e.g., positive versus negative) of ITs (e.g., Lu and Ramamurthy 2011) for different business conditions.

Second, our nomological model can be further extended by investigating alternative moderators or mediators. In the literature, several other capabilities (along with IT resources and capabilities) have been discussed with respect to their impact on firm performance or higher-order dynamic capabilities. For

example, some characteristics of the top management team, such as entrepreneurial insight, are considered important facilitating factors of IT-enabled competitive success, particularly for firms operating in a dynamic environment (Chen et al. 2010, Sambamurthy et al. 2003). Tallon (2008) suggests investigating the mediating role of technical IT capabilities, such as flexibility and adaptability of IT infrastructure, between IT management capabilities and organizational agility. Therefore, future research should further examine how other potential capabilities and conditions may interact with IT ambidexterity to enable organizational agility or other types of dynamic capabilities (e.g., new product development capability), especially under the different environmental conditions.

Next, IT ambidexterity needs to be further investigated to better understand this emergent capability and its impacts within firms. In particular, instead of examining IT ambidexterity as a combined capability, IT exploration and IT exploitation capabilities can be segregated and examined in terms of the distinct values of their various combinations under different business conditions. Understanding the impacts of firms' distinct combinations of IT exploration and IT exploitation capabilities especially under the different

environmental conditions may be an interesting and important agenda for future research in this area. Appendix D shows more details on the potential promise of this future research by showing some preliminary analyses based on our data.

Conclusion

We set forth to answer two unanswered questions in the literature: (1) How does IT ambidexterity enhance organizational agility within a firm? (2) How does the firm's business environment affect the capability-building processes between IT ambidexterity and organizational agility? In answering these questions, we developed and tested a nomological model of the relationships among IT ambidexterity, operational ambidexterity, organizational agility, and environmental dynamism. By clearly identifying the underlying mechanism and boundary conditions of the IT-enabled capability building process, our study serves as a building block for future research in this important and growing area of research. By empirically validating the relationship between IT ambidexterity and operational ambidexterity, our study demonstrates to IT executives the importance of a balance between IT exploration and exploitation.

Appendix A. Summary of Selected Research on the IT Impact on Organizational Agility

Research	Context/Research focus	Theoretical lens	Methodology	Key arguments and findings
Sambamurthy et al. (2003)	IT impact on agility through digital options	Capability-building perspective	Theoretical development	IT resources provide digital process and knowledge options as enablers of agility. Entrepreneurial alertness moderates the links between these enablers, agility and firm performance.
Tallon (2008)	Impact of IT capabilities on business process agility	Contingency theory	Empirical validation using field survey data	Managerial and technical IT capabilities impact agility. Furthermore, in stable environments, technical IT capabilities are more significant, whereas in a dynamic setting, managerial IT capabilities are more important.
Pavlou and El Sawy (2010) ^a	IT impact on new product development capabilities	Dynamic capability view and contingency perspective	Empirical validation using field survey data	Different roles of improvisational and dynamic capabilities and the variation in their significance across different levels of environmental dynamism.

Appendix A. (Continued)

Research	Context/Research focus	Theoretical lens	Methodology	Key arguments and findings
Lu and Ramamurthy (2011)	IT impact on market capitalizing agility and operational adjustment agility	Complementarity perspective	Empirical validation using field survey data	IT capability and IT spending have a joint effect on operational adjustment agility, but not on market capitalizing agility. Though more IT spending itself does not enable agility, its complementarity with IT capabilities has a significant effect on agility.
Mithas et al. (2011) ^a	Impact of information management capability on firm performance	IT capability and resource-based view	Empirical validation using archival data	IT capabilities enhance organizational performance through the mediating effects of specific business capabilities, i.e., customer, process, and performance management capabilities.
Tallon and Pinsonneault (2011)	Impact of strategic IT alignment on firm performance through agility enhancement	Strategic IT alignment	Empirical validation using field survey data	Agility fully mediates the effects of strategic IT alignment on firm performance. These effects are stronger in highly volatile environments.
Nazir and Pinsonneault (2012)	IT impact on sensing and responding capabilities through knowledge processes	Electronic integration perspective	Theoretical development	External and internal electronic integrations selectively lead to sensing and responding capabilities, respectively, through knowledge exploration, exploitation, and process coupling.
Chakravarthy et al. (2013)	Enabling and facilitating the effects of IT capabilities on agility in the presence of multiple contingencies	Theory of multiple contingencies	Empirical validation using field survey data	IT capabilities enhance organizational agility through an enabling role, but they also moderate the effects of agility on firm performance through facilitating roles. Environmental dynamism further moderates both of these effects.

^aAlthough these two studies do not directly address the issue of agility, they are included in this table due to their relevance to our research.

Appendix B. Measurement Items for Survey Questionnaires

Organizational agility. Relative to other firms in your industry, please indicate the ability of your firm to:

Proactiveness	Anticipate new business opportunities
	Seek new business opportunities
	Seek novel approaches to future market needs
Radicalness	Seek high-risk projects with chances of high return
	Support business experimentation despite uncertain returns
	Commit resources to radical changes that can potentially transform markets and competition
Responsiveness	Rapidly react to emerging opportunities in customer needs
	Rapidly react to emerging opportunities in markets
	Rapidly react to emerging environmental opportunities (e.g., new regulations, globalization)
Adaptiveness	Adapt existing business models
	Adapt existing business processes
	Quickly adopt best practices used by others

IT ambidexterity. Relative to other firms in your industry, please indicate the ability of your IT unit(s) to:

IT exploration capability	Acquire new IT resources (e.g., new generation of IT architecture, potential IT applications, critical IT skills)
	Experiment with new IT resources
	Experiment with new IT management practices
IT exploitation capability	Reuse existing IT components, such as hardware and network resources
	Reuse existing IT applications and services
	Reuse existing IT skills

Operational ambidexterity. Relative to other firms in your industry, please indicate the ability of your operational unit(s) to:

Operational exploration capability	Implement extensive innovations in business operations (e.g., product/service development and production, supply chain management, customer delivery, employee management)
	Implement radical innovations in business operations
	Implement operational innovations that are difficult to replicate by other firms
Operational exploitation capability	Reduce the cost of existing business operations
	Improve the cycle time of existing business operations
	Improve the efficiency of existing business operations

Environmental dynamism. Please indicate your degree of agreement about your market and competitive environment:

Environmental dynamism	The actions of competitors in your major markets are changing rapidly.
	Technological changes in your industry are rapid.
	Customers' product/service preferences change rapidly.

Appendix C. Evidence of Measurement Reliability and Validity of the Original Measures

Constructs	Items	Mean (SD)	Reliability analysis		Exploratory factor analysis								
			Item reliability (factor loadings)	Construct reliability (Cronbach's alpha)	PRO	RAD	RES	ADP	OERC	OEIC	IERC	IEIC	EVD
Proactiveness (PRO)	PRO1	5.14 (0.82)	0.91	0.89	0.80	0.25	0.26	0.13	0.08	0.14	0.05	−0.03	0.07
	PRO2	5.28 (0.80)	0.93		0.81	0.23	0.25	0.19	0.07	0.16	0.10	−0.04	0.04
	PRO3	5.29 (0.77)	0.87		0.66	0.14	0.40	0.17	0.28	0.19	0.01	0.11	0.05
Radicalness (RAD)	RAD1	3.88 (1.21)	0.91	0.88	0.22	0.75	0.11	0.22	0.28	0.12	0.08	0.01	0.23
	RAD2	3.75 (1.25)	0.93		0.21	0.80	0.18	0.17	0.13	0.21	0.14	−0.02	0.18
	RAD3	4.10 (1.13)	0.86		0.16	0.70	0.23	0.24	0.18	0.08	0.03	0.18	0.24
Responsiveness (RES)	RES1	5.22 (0.81)	0.93	0.89	0.39	0.13	0.73	0.09	0.25	0.19	0.06	0.12	0.17
	RES2	5.19 (0.79)	0.93		0.41	0.07	0.71	0.20	0.25	0.12	0.10	0.05	0.20
	RES3	4.92 (0.85)	0.87		0.16	0.29	0.74	0.15	0.16	0.14	0.10	0.13	0.11

Appendix C. (Continued)

Constructs	Items	Mean (SD)	Reliability analysis		Exploratory factor analysis								
			Item reliability (factor loadings)	Construct reliability (Cronbach's alpha)									
					PRO	RAD	RES	ADP	OERC	OEIC	IERC	IEIC	EVD
Adaptiveness (ADP)	ADP1	4.23 (1.00)	0.92	0.87	0.22	0.25	0.05	0.76	0.21	0.17	0.23	0.01	0.17
	ADP2	4.34 (0.96)	0.94		0.16	0.20	0.12	0.78	0.22	0.19	0.21	0.04	0.14
	ADP3	4.58 (1.07)	0.83		0.10	0.17	0.23	0.79	0.00	0.18	0.05	0.07	0.04
Operational exploration capability (OERC)	OIC1	4.94 (0.87)	0.88	0.87	0.15	0.20	0.21	0.06	0.78	0.22	0.03	0.04	0.15
	OIC2	4.37 (0.98)	0.92		0.05	0.19	0.22	0.07	0.83	0.15	0.10	0.10	0.16
	OIC3	4.13 (1.19)	0.89		0.16	0.16	0.15	0.31	0.73	0.15	0.15	0.02	0.22
Operational exploitation capability (OEIC)	PIC1	4.80 (0.82)	0.85	0.87	0.38	0.01	0.03	0.08	0.11	0.78	0.17	0.12	0.07
	PIC2	4.61 (0.91)	0.91		−0.01	0.19	0.22	0.19	0.20	0.78	0.18	0.22	0.12
	PIC3	4.79 (0.88)	0.91		0.09	0.17	0.16	0.22	0.17	0.82	0.07	0.04	0.17
IT exploration capability (IERC)	ERC1	4.64 (0.91)	0.86	0.87	0.10	0.01	0.13	0.06	0.08	0.11	0.78	0.25	0.15
	ERC2	4.47 (0.96)	0.91		0.04	0.11	0.03	0.09	0.09	0.16	0.83	0.25	0.11
	ERC3	4.39 (0.93)	0.89		0.02	0.13	0.07	0.26	0.06	0.11	0.79	0.19	0.10
IT exploitation capability (IEIC)	EIC1	5.15 (0.96)	0.89	0.87	−0.03	0.12	0.13	0.12	−0.03	0.11	0.18	0.86	0.02
	EIC2	5.00 (0.88)	0.92		−0.01	0.04	0.09	0.00	0.09	0.11	0.17	0.87	0.14
	EIC3	5.19 (0.92)	0.86		0.07	−0.03	0.02	0.00	0.09	0.09	0.31	0.78	0.11
Environmental dynamism (EVD)	PET1	4.67 (1.31)	0.83	0.78	0.22	0.33	−0.03	0.25	0.21	0.16	0.13	0.10	0.63
	PET2	4.42 (1.51)	0.86		0.10	0.26	0.17	0.23	0.18	−0.03	0.15	0.07	0.73
	PET3	4.03 (1.50)	0.81		−0.09	0.06	0.17	−0.08	0.09	0.19	0.10	0.10	0.85

Note. Through the exploratory factor analysis, all items loaded more highly on the intended construct than on any other constructs.

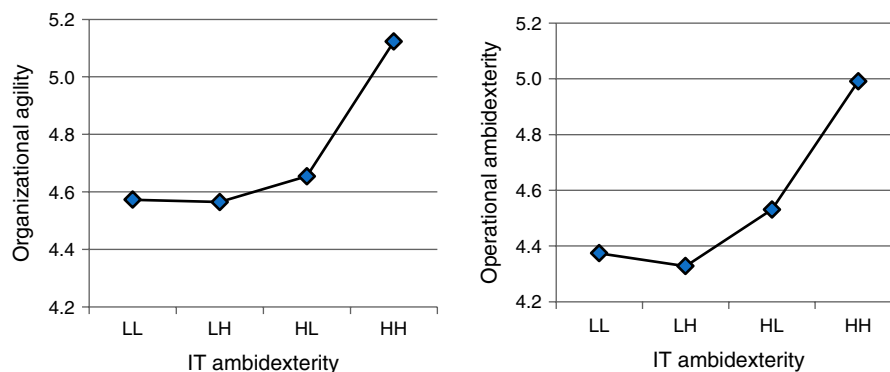
Appendix D. Different Combinations of IT Exploration and Exploitation

The samples used for this research can be categorized into four different combinations of low–low ($n = 63$), low–high ($n = 20$), high–low ($n = 42$), and high–high ($n = 53$) for IT exploration and IT exploitation capabilities using the averages of these capabilities, respectively. As shown in Figure D.1, the mean-value comparisons among the four groups using one-way ANOVA tests provide interesting findings.

For both organizational agility and operational ambidexterity, only the high–high groups (high IT exploration and

high IT exploitation) have significant differences ($p < 0.01$) from all other combinational groups, whereas the three other groups are not significantly different in terms of their mean values. Interestingly, however, the high–low groups (high IT exploration and low IT exploitation) show higher mean values of both organizational agility and operational ambidexterity than the low–high (low IT exploration and high IT exploitation) and the low–low groups (low IT exploration and low-high IT exploitation), although their differences were not significant. Moreover, the mean values of the low–high groups are slightly lower than are those of the low–low groups.

Figure D.1. Plots of Means for Different Combinations of IT Exploration and Exploitation



Notes. LL, Low IT exploration and low IT exploitation ($n = 63$); LH, low IT exploration and high IT exploitation ($n = 20$); HL, high IT exploration and low IT exploitation ($n = 42$); HH, high IT exploration and high IT exploitation ($n = 53$).

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